

M4350 - Create VLAN flow via CLI

Command:

vlan

M4250

M4350

This command creates a new VLAN and assigns it an ID. The ID is a valid VLAN identification number (ID 1 is reserved for the default VLAN). The VLAN number is in the range 2–4093.

Format	vlan number
--------	-------------

Mode	Global Config
------	---------------

Flow:

CLI Command Registration:

ewsCliAddNode():

- Registers a CLI command.
- Important Inputs:
 - Callback Function: commandVlanSwDev()
 - Command String: "vlan"
 - "no" command support.

Management Layer:

CLI Handler:

- commandVlanSwDev()
- cliParseRangeInput(): Parses Input and prepares a list of VLANs to be created.
- Creates a mask from the list: cliVlanList
- Calls usmDbVlanCreateMaskSet() with vlan mask as argument.

USMDB:

- usmDbVlanCreateMaskSet()
- Calls dot1qVlanCreateMask() with the mask

Application Layer:

- dot1qVlanCreateMask():
 - Inside a OSAPI write lock, for each VLAN set in mask:
 - Checks if VLAN already exists - dot1qVlanCheckValid()
 - Check is VLAN is configurable - dot1qVlanIsConfigurable()
 - If both fail, create configuration entry for the VLAN.
 - Notify registered users with VLAN_NAME_CHANGE_NOTIFY event.
 - Increment VLAN count.
 - dot1qIssueCmd:
 - Posts the VLAN mask in a message format to dot1q task.
 - Event: dot1q_vlan_create_static_mask
 - Uses osapiMessageSend for asynchronous processing.
- dot1qDispatch():
 - Dot1q event processor.
- dot1qVlanCreateMaskProcess()
 - Inside a OSAPI write lock, Loops through VLAN mask and creates entry(vlanEntryAdd) in the VLAN tree.
 - Calls dtlDot1qCreateMask().
 - Notify registered users with VLAN_ADD_NOTIFY using vlanNotifyRegisteredUsers()
 - Enable VLAN statistics - dtlEnableVlanStats()

Device Transformation Layer (DTL):

- dtlDot1qCreateMask():
 - Calls dapiCtl() with DAPI command DAPI_CMD_QVLAN_VLAN_LIST_CREATE.

DAPI:

- dapiCtl():
 - DAPI's database dapi_g, has a list of port specific function pointers(cmdTable) for each DAPI command.
 - hapiBroadQvlanVlanListCreate is the function pointer for DAPI_CMD_QVLAN_VLAN_LIST_CREATE
 - Init for this is done in hapiBroadL2VlanPortInit()

HAPI:

hapiBroadQvlanVlanListCreate



usl_bcmx_vlan_bulk_configure



l7_rpc_client_vlan_bulk_configure



l7_rpc_vlan_bulk_configure



hpcHardwareRpc



hpcHardwareRpcDispatch



l7_rpc_server_vlan_bulk_update (RPC Server/Handler - Registered)



usl_bcm_vlan_update



bcm_vlan_create() (Broadcom SDK API)